Industrial Internship Report on

"Full-Stack Development"

Prepared by

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time. My project was "Banking Information System", which is a simple Java-based application developed during my Full Stack Development internship. The system performs basic banking operations such as account creation, deposit, withdrawal, and balance checking. This internship helped me understand how programming concepts are applied in real-world situations. I also improved my knowledge of Java, object-oriented programming, and basic system design. Overall, this internship was a great learning experience and helped me build confidence in my technical skills. This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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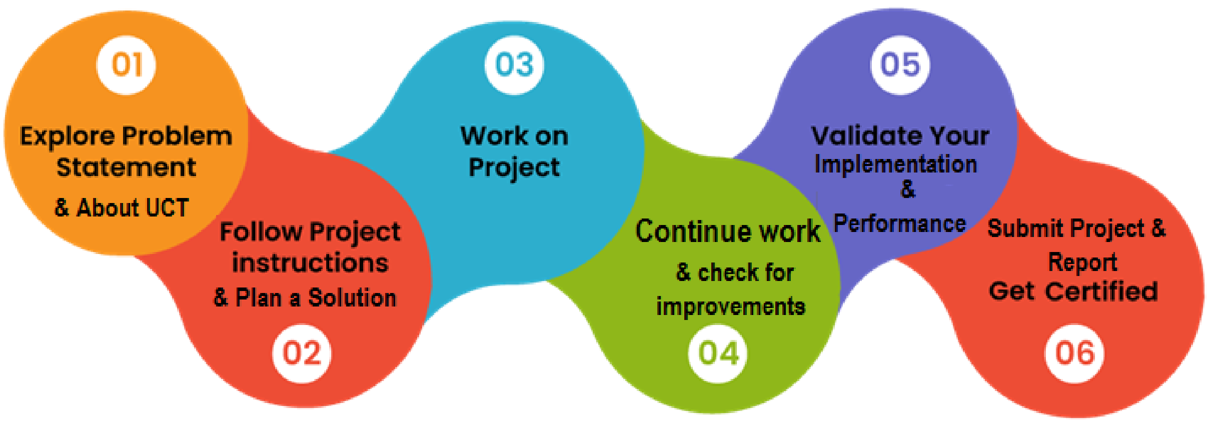
1 Preface

This report presents the summary of my 6-week internship in Full Stack Development conducted by upskill Campus in collaboration with UniConverge Technologies Pvt Ltd.

During this internship, I worked on a project titled “Banking Information System”, where I applied my knowledge of Java programming to develop a basic banking application. This project helped me understand how theoretical concepts can be used in practical applications.



This internship was very useful for my career development as it improved my programming skills and problem-solving ability. It also gave me an idea of how real-world systems are designed and developed.

I would like to thank upskill Campus, UCT, and all the mentors for providing me this opportunity and guiding me throughout the internship.

I would also suggest my juniors to take such internships seriously, as they help in gaining practical knowledge and building a strong foundation for future career growth

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

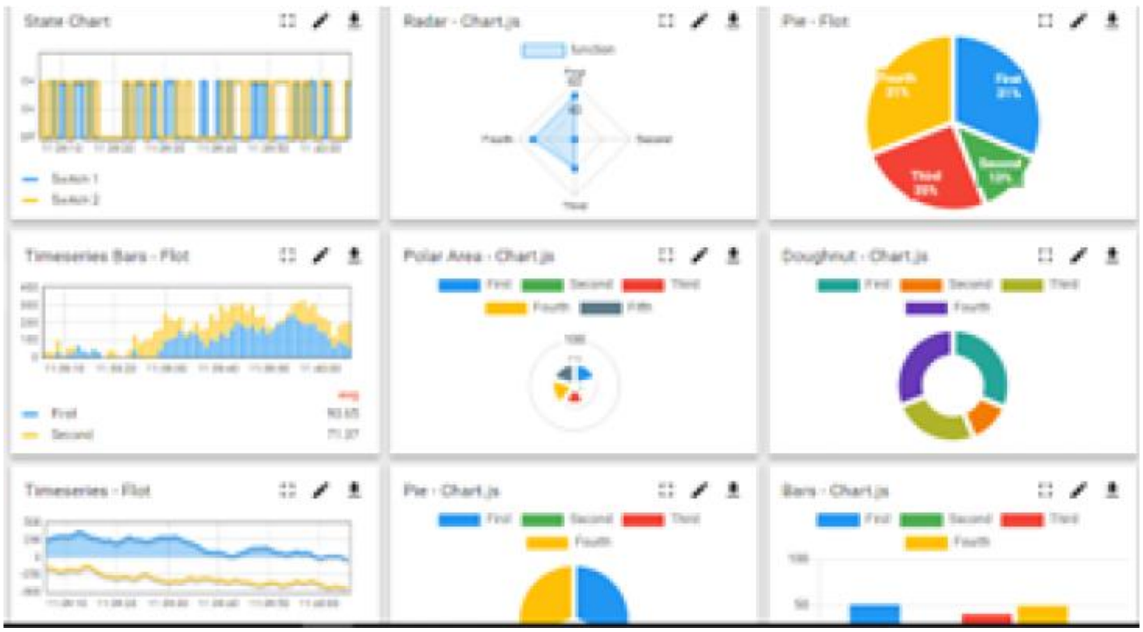
i. UCT IoT Platform ( )

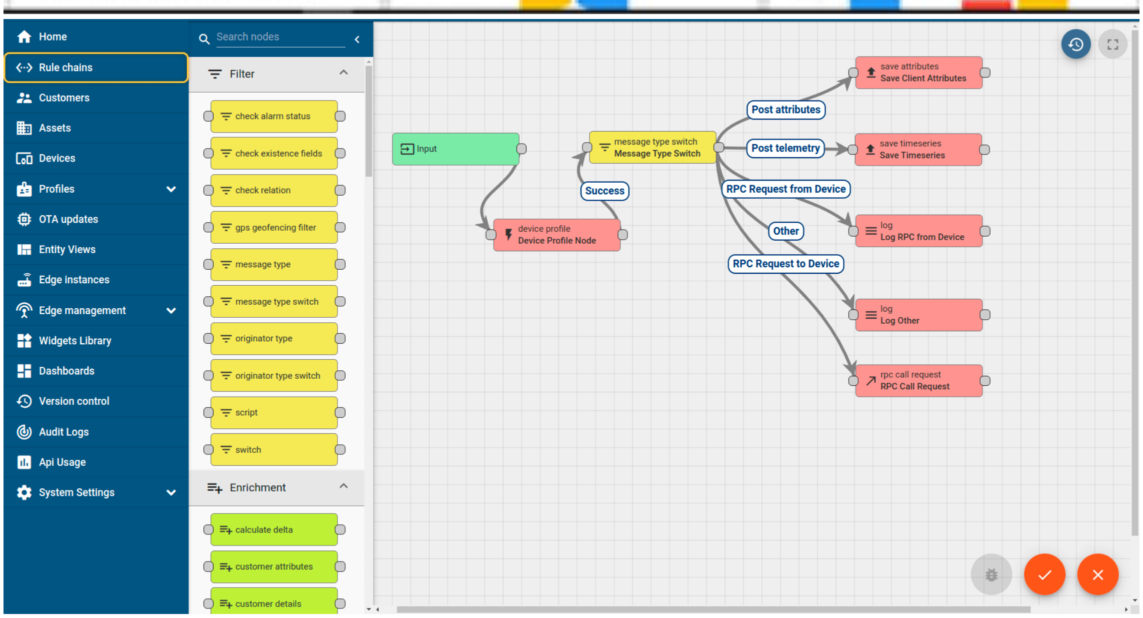
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

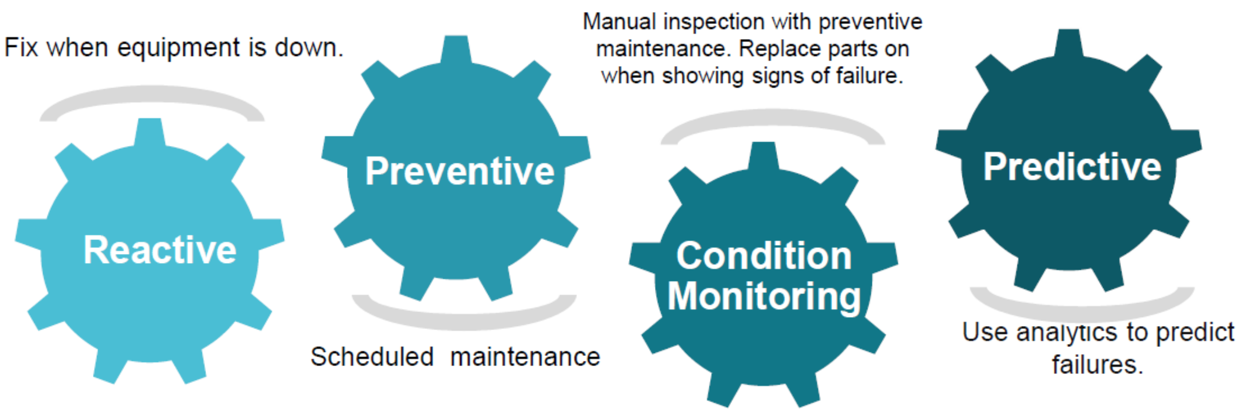
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

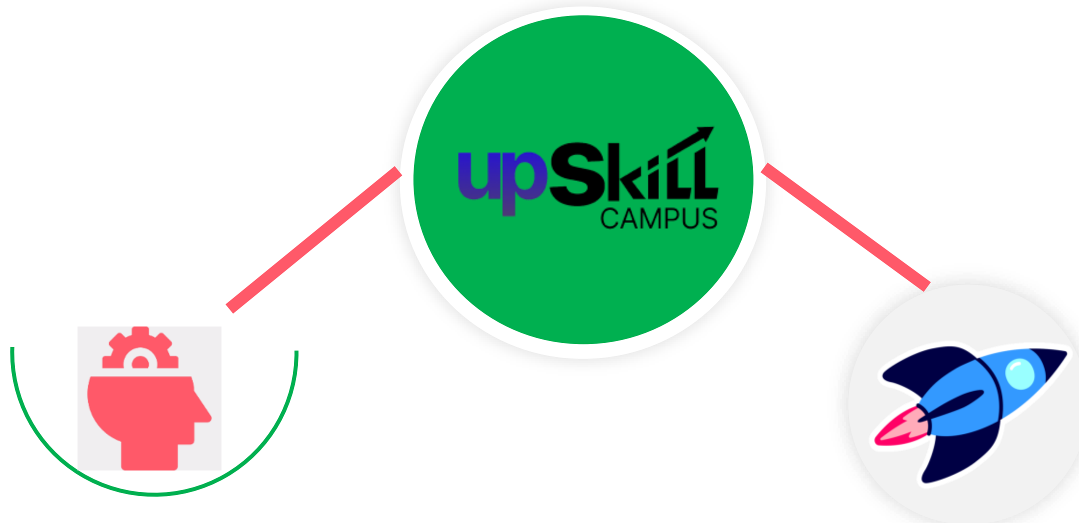
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/

Career growth/upskilling
- Interview Preparation and skill building
- upskilling Courses
- Skill Assessment
- Profile building

Professional networking
- Alumni Connections
- Mentorship
- Discussion/QA forum

Collaboration platform
- Project collaboration
- Discussion forum
- Tech updates

Job/internship platform
- Job portal
- Internship portal
- Freelancing projects

2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

☛ get practical experience of working in the industry.

☛ to solve real world problems.

☛ to have improved job prospects.

☛ to have Improved understanding of our field and its applications.

☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1] Java Programming Study Material provided during internship
[2] Official documentation of Java (Oracle)
[3] Learning resources from upskill Campus portal

2.6 Glossary

Terms	Acronym
Object-Oriented Programming	O**OP**
Java Database Connectivity	J**DBC**
HyperText Markup Language	H**TML**
Structured Query Language	S**QL**
Application Programming Interface	A**PI**

3 Problem Statement

In the assigned problem statement

In the current scenario, managing banking operations manually can be time-consuming and prone to errors.

The objective of this project is to develop a simple Banking Information System using Java that can handle basic operations like account creation, deposit, withdrawal, and balance checking in an efficient and organized way.

4 Existing and Proposed solution

Existing System:

The existing system mainly relies on manual record keeping, which is slow and can lead to errors. Managing customer data manually is not efficient and requires more time and effort.

Proposed System:

The proposed system is a Java-based application that automates basic banking operations. It allows users to perform operations quickly and accurately without manual intervention.

Value Addition:

This system reduces human errors, saves time, and makes data handling easier and more efficient.

4.1 Code submission (Github link: https://github.com/prateek28bc053-tech/upskillcampus/blob/main/BankingInformationSystem.java

4.2 Report submission (Github link) : https://github.com/prateek28bc053-tech/upskillcampus/blob/main/BankingInformationSystem_PrateekDubey_USC_UCT.pdf

5 Proposed Design/ Model

The system is designed in a simple way where the user provides input through the console. The Java program processes the input using predefined methods and displays the result accordingly.

The overall flow of the system is:
User → Input → Java Processing → Output

5.1 High Level Diagram (if applicable)

Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)

5.3 Interfaces (if applicable)

Update with Block Diagrams, Data flow, protocols, FLOW Charts, State Machines, Memory Buffer Management.

6 Performance Test

The system was tested for different operations such as account creation, deposit, withdrawal, and balance checking.

The application performs efficiently with minimal memory usage and provides accurate results. It can handle multiple operations smoothly without errors.

6.1 Test Plan/ Test Cases

6.2 Test Procedure

6.3 Performance Outcome

7 My learnings

During this internship, I learned important concepts of Java programming, including object-oriented programming, inheritance, polymorphism, and abstraction.

I also gained knowledge of Java 8 features and basic database connectivity using JDBC. In addition, I understood the fundamentals of full stack development, including frontend, backend, and database interaction.

This internship improved my logical thinking, coding skills, and overall understanding of software development.

8 Future work scope

In the future, this project can be enhanced by developing a web-based interface using HTML, CSS, and JavaScript.

Additional features such as user login, database integration, and online transaction support can be added to make the system more advanced and suitable for real-world applications.